

Ben Jacobsen

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Research Experience

- **Private Learning (ongoing)** Madison, WI
Graduate Research Assistant Dec 2022 - present
 - Designing, analyzing, and evaluating practical machine learning systems that provably satisfy differential privacy while maintaining high performance. We are particularly interested in *online learning*, with applications to real-time generation of private synthetic data.
- **MetroSets** Tucson, AZ
Undergraduate Research Assistant Sep 2019 - Jan 2021
 - Designed and implemented a novel system for set visualization using the metro map metaphor, with sets drawn as subway lines and elements drawn as stations. Subsequently conducted a controlled user study which found that MetroSets outperformed other publicly available tools.
- **Authorship Attribution** Tucson, AZ
Honors Thesis May 2020 - Jul 2021
 - Studied the robustness of machine learning systems designed to classify the author of a binary file. Designed and implemented an adversarial attack on state-of-the-art tools by using non-standard compiler optimizations to obscure an author's stylistic fingerprint.

Publications

Jacobsen, B., Wallinger, M., Kobourov, S., and Nollenburg, M. (2020). *MetroSets: Visualizing Sets as Metro Maps*. IEEE Transactions on Visualization and Computer Graphics.

Wallinger, M., Jacobsen, B., Kobourov, S., and Nollenburg, M. (2021). *On the Readability of Abstract Set Visualizations*. IEEE Transactions on Visualization and Computer Graphics.

Jacobsen, B., Rahaman, S., and Debray, S. (2021) *Optimization to the Rescue: Evading Binary Code Stylometry with Adversarial Use of Code Optimizations*. Proceedings of the 2021 Research on offensive and defensive techniques in the Context of Man At The End (MATE) Attacks.

Education

- **University of Wisconsin - Madison** Madison, WI
PhD Student, Computer Science Expected 2027
- **University of Wisconsin - Madison** Madison, WI
M.S., Computer Science May, 2024
- **University of Arizona (Honors College)** Tucson, AZ
B.A., Mathematics, minor in Computer Science, Summa Cum Laude May 2021
- **Central New Mexico Community College** Albuquerque, NM
A.S. with Highest Honors, Mathematical Sciences August 2019

Work Experience

- **Graduate Teaching Assistant**
University of Wisconsin - Madison *Aug 2021 - present*
- **Undergraduate Research Assistant**
University of Arizona *Sep 2019 - May 2021*
- **Research Data Support Specialist**
University of Arizona *June 2021 - August 2021*
- **Programming Tutor**
Private *June 2020 - August 2021*
- **Software Consultant**
Fyberloom Inc. *June 2021 - July 2021*

Awards and Honors

UA College of Science Excellence in Undergraduate Research Award (Overall Winner)	2021
CRA Outstanding Undergraduate Researcher Honorable Mention	2020
Phi Beta Kappa Honors Society	2021-Present
Phi Theta Kappa Honors Society	2018-Present
Phi Theta Kappa Transfer Scholarship (\$24,000)	2019-2020
Galileo Circle Scholarship (\$1,000)	2020
Best Poster GD2020	2020

Skills

Programming Languages: Python, Java, MATLAB

Natural Languages: English, Mandarin (intermediate)

Operating Systems: Linux (Arch, Ubuntu), Windows

Tools: L^AT_EX, Git, Docker, VirtualBox, NumPy, Pandas, Matplotlib, PyTorch, TensorFlow

Miscellaneous: creative problem solving, public speaking, formal and persuasive writing